

This is the General Round. You have 40 minutes to complete 20 problems.

This is an individual round.

There is no penalty for guessing. Answer submissions are only valid if they are written on the answer sheet below.

First and last name			
p	Answer	p	Answer
1		11	
2		12	
3		13	
4		14	
5		15	
6		16	
7		17	
8		18	
9		19	
10		20	

Problem 1. The probability that Angie has an umbrella given that it is raining is 30%. The probability that it rains is 60%. What is the probability that Angie has an umbrella, and it is raining? Express your answer as a percentage.

Problem 2. Find the number of positive integers k such that $\frac{30}{k}$ is an integer.

Problem 3. Rectangle $ABCD$ has side length $AB = 8$. Let E be the intersection of AC and BD . Given that $AE = 5$, find $\frac{[BCD]}{[AED]}$. Note that $[BCD]$ and $[AED]$ represent the areas of triangles BCD and AED , respectively.

Problem 4. Sneha, Evan, and Taison are comparing their ages. Sneha says, “My age seven years ago was exactly half of Taison’s age seven years ago.” Taison replies, “My age right now is the average of Sneha and Evan’s ages.” Finally, Evan says “I’m ten years older than Sneha.” What is the sum of their ages?

Problem 5. Find the smallest prime factor of $4^5 - 7^2$.

Problem 6. Let Ω be a circle with center O , and let $\overline{BC}$ be a diameter of Ω . Let A be a point on Ω such that $\angle ABC = 45^\circ$. If the area of $\triangle AOC$ is 67, determine the area of Ω .

Problem 7. Sam eats lunch at a cafeteria. He has three options for a main dish, four options for a side dish, and five options for dessert. If Sam wants to get one main dish, one side dish, and two desserts, how many distinct lunches could Sam eat? (It is not allowed to get the same dessert twice, and the order of desserts does not matter.)

Problem 8. Austin and Jeff run to an ice cream store at their own constant speeds. The ice cream store is a mile away from where Austin and Jeff start. Jeff gets there first, and spends fifteen minutes ordering and eating his ice cream. He then gets bored of waiting, and begins running back towards Austin. He meets Austin halfway between the ice cream store and their starting location. If Jeff runs at 15 mph, what is Austin’s speed in mph?

Problem 9. Let a and b be positive numbers. Given that $a^2b^2 = 169$ and $3a + 3b = 27$, compute $(a - b)^2$.

Problem 10. Let $ABCD$ be a trapezoid with $AB \parallel CD$. Given that $AB = 17$, $BC = 25$, $CD = 45$, and $DA = 17$, compute the area of $ABCD$.

Problem 11. Austin, Eric, Sam, Sneha, and Ryan are waiting in line to use the communal library computer to search for the books they want. Austin’s search takes 4 minutes to complete, Eric’s takes 8 minutes, Sam’s takes 2 minutes, Sneha’s takes 6 minutes, and Ryan’s takes 3 minutes. If the five people are randomly arranged in line to the computer, and only one person can use it at a time, what is the expected number of minutes that will pass until Ryan completes his search?

Problem 12. The positive integer n satisfies the condition that $n^2 + n$ has 5 prime divisors and 32 positive divisors. Find the smallest possible value of n .

Problem 13. Bob wants to color each of the numbers in the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ either red, green, or blue, so that there are at least three elements of each color, including one element which is a multiple of three. How many such colorings are possible?

Problem 14. For positive integers a , b , c , and d , we have

$$\frac{a}{2} - \frac{b}{3} + \frac{c}{5} - \frac{d}{7} = \frac{1}{210}.$$

Find the smallest possible value of $1000a + 100b + 10c + d$.

Problem 15. Let r_1 and r_2 denote the roots of the polynomial $x^2 - 7x + 1$. Compute $r_1\sqrt{r_1} + r_2\sqrt{r_2}$.

Problem 16. Let $ABCDEF$ be a regular hexagon of side length 12 and let ω be its circumcircle. Point P lies on minor arc AB of ω such that $PE = 13\sqrt{3}$. Compute $PF \cdot PD$.

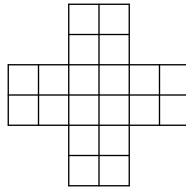
Problem 17. Let f be a function such that $f(1) = 2$ and $f(n) = f(n-1)^2$ for all $n \geq 2$. Find the remainder when $f(f(f(3)))$ is divided by $2^7 - 1 = 127$.

Problem 18. Let S be the value of the infinite sum

$$\sum_{n=2025}^{\infty} \frac{n^2 - n - 1}{n!} = \frac{2025^2 - 2025 - 1}{2025!} + \frac{2026^2 - 2026 - 1}{2026!} + \cdots.$$

The number $N = 2025! \cdot S$ is an integer. Compute the number of positive divisors of N .

Problem 19. Consider the board below, consisting of a 6×6 square with the corner 2×2 squares removed. How many ways are there to completely tile the board using only 1×2 and 2×1 pieces?



Problem 20. Let $ABCDE$ be a pentagon inscribed in a circle with $AB = BC$. Let P be the intersection of chords AD and BE , Q be the intersection of chords BD and CE , and R be the intersection of chords BD and AC . Given that $AP = 3$, $PD = 9$, and $AR = 8$, determine the length of $\overline{PQ}$.